

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 03)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

1. Write and run a Python program with comments (named **Lab3_1.py**) that
 - Let user input a string, and display it to the terminal
 - Let user input an integer, and display it to the terminal
 - Let user input a floating point number, and display it to the terminal

Sample Program Output

```
Input a string, followed by <ENTER> key: We Love ICP
String < We Love ICP > is entered!
Input an Integer, followed by <ENTER> key: 4020
Integer < 4020 > is entered!
Input a float, followed by <ENTER> key: 40.2
Float number < 40.2 > is entered!

..... By YourName; ICP 4020; 2021 .....
```

←--- User Input,
with <ENTER>

2. Write and run a Python program that calculates area of a circle and surface area of a sphere (named **Lab3_2.py**):

- Use and import the **math** module to finish this section.
- Display the following prompt: **"Enter the radius of a circle: "**
- Accept input from console terminal a value for the radius (e.g. 3.3)
- Display the value of π (**pi**) of the **math** module on screen
- Calculate and display the **area of a circle** with 2 decimal digits.

Hint: area of a circle = πr^2

- Use the same value of radius above, calculate and display the **surface area of the sphere** with 2 decimal digits.

Hint: surface area of a sphere = $4\pi r^2$

Sample Program Output

```
Enter the radius of a circle: 3.3
Pi is 3.141592653589793
The area of the circle = 34.21
The surface area of the sphere = 136.85

..... By YourName; ICP 4020; 2021 .....
```

←--- User Input,
with <ENTER>

3. Write and run a Python program (named **Lab3_3.py**) that calculates the distance between two points where **point coordinates are input by user**.

- Use (and import) the **math** module.
- Display a prompt message.
- Let user input parameters of point coordinates and store in the variables: x_1 , y_1 , x_2 , and y_2 .
- Calculate and display the **distance** with 6 decimal digits.

Given: Equation of distance between two points having coordinates (x_1, y_1) and (x_2, y_2) is

$$\text{distance} = \text{sqrt} ([x_1 - x_2]^2 + [y_1 - y_2]^2)$$

Sample Program Output

```
Enter 2 points (x1, y1) and (x2, y2) for calculating their distance:
Enter x1: 7
Enter y1: 12
Enter x2: 3
Enter y2: 9
The distance between two points is 5.000000
..... By YourName; ICP 4020; 2021 .....
```

~ END ~